KEYENCE

D67GB

Auto-ID Technical Guide

SR-X300/X100/2000/1000
SR-EC1
OMRON NJ-100 SERIES

Connection Guide: EtherCAT Communication

This manual explains how to use EtherCAT communication to connect to the code reader and the Omron PLC via SR-EC1

Table of Contents

Chapter 1 EQUIPMENT USED
Chapter 2 CONNECTION METHODS
Chapter 3 CONFIGURING OMRON MASTER SETTINGS
Chapter 4 EXECUTING LADDER PROGRAMMES AND MONITORING READING DATA

Chapter 1 EQUIPMENT USED

Prepare the following equipment.

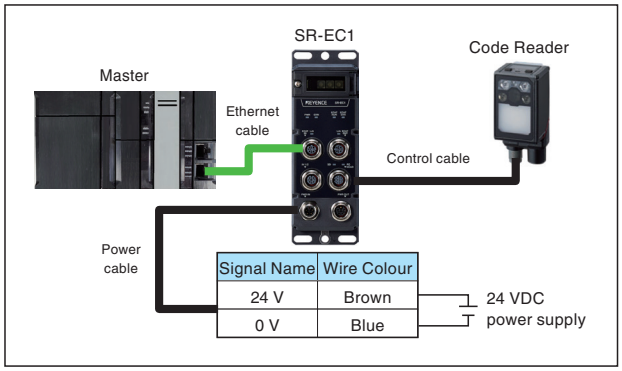
CPU module	Code Reader
SR-EC1	Ethernet cable OP-87230 (2 m)
Control cable*1	Power cable OP-88782 (2m)

*1 OP-88776(2m) (SR-X300/X100)
OP-88779(2m) (SR-2000/1000)

Chapter 2 CONNECTION METHODS

Overall connection diagram

Connect the master and the SR-EC1 as shown below.

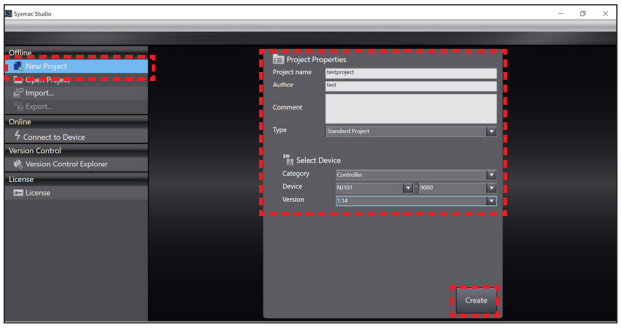


Chapter 3 CONFIGURING ETHERCAT MASTER SETTINGS

* Sysmac Studio V1.2 or later is required to configure EtherCAT settings.

1 Use Sysmac Studio to create a new project.

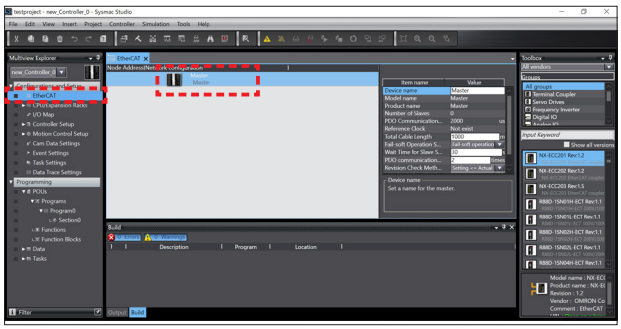
Start Sysmac studio and click "New Project".
Enter "Project name", "Category", "Device", "Version" and Click "Create".



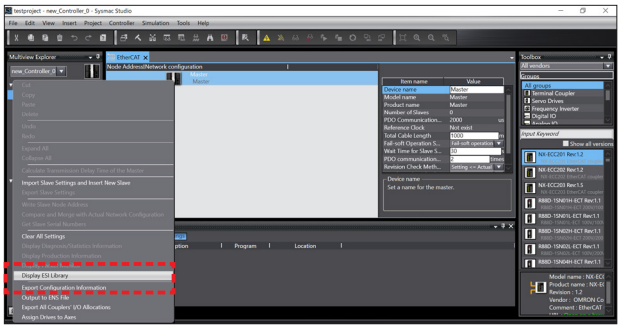
2 Register the ESI file (definitions file for SR-EC1 EtherCAT) for the communication unit.

You can download the Tag Description ESI file on the WEB (barcodereader.com).

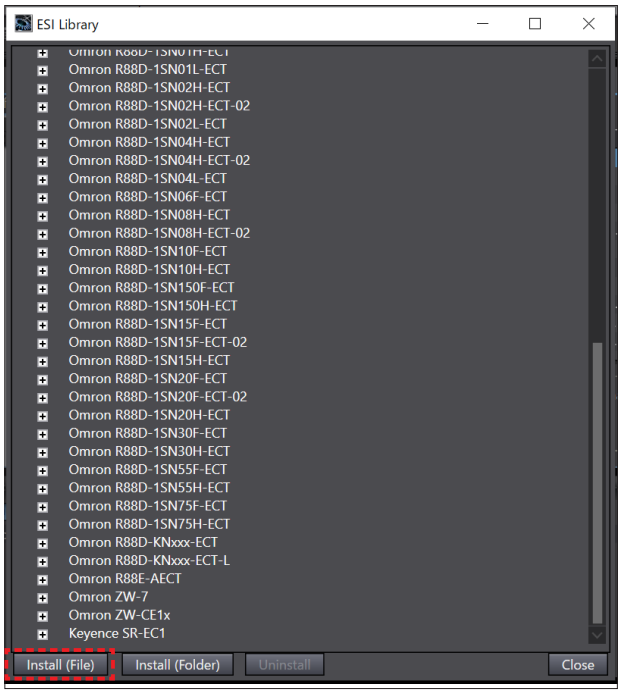
Click "EtherCAT" in the Multiview Explorer and right click the Master.



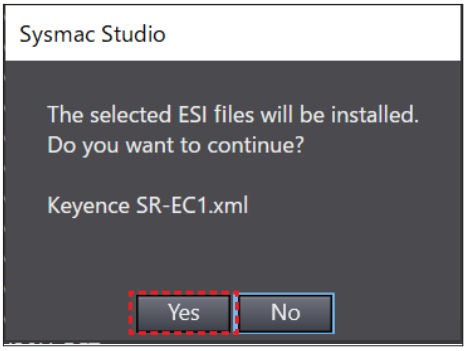
Select "Display ESI Library".



Click "Install (File)" and select ESI which you downloaded.

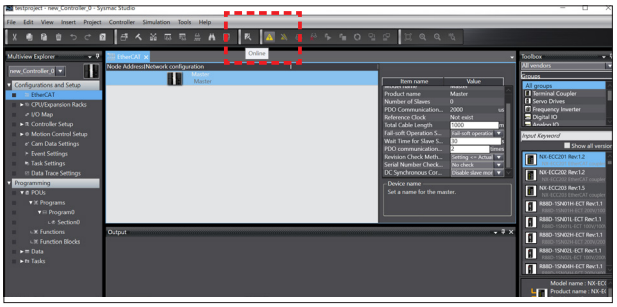


Click "Yes" button in the following pop-up window.

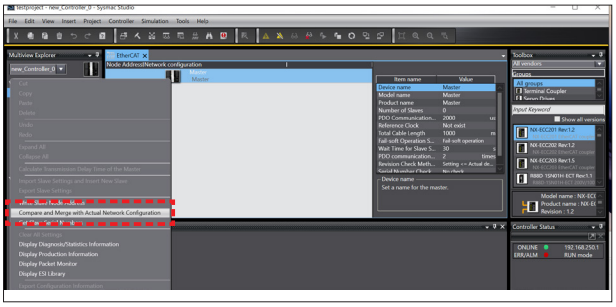


3 Add SR-EC1 to the Node Address Network configuration.

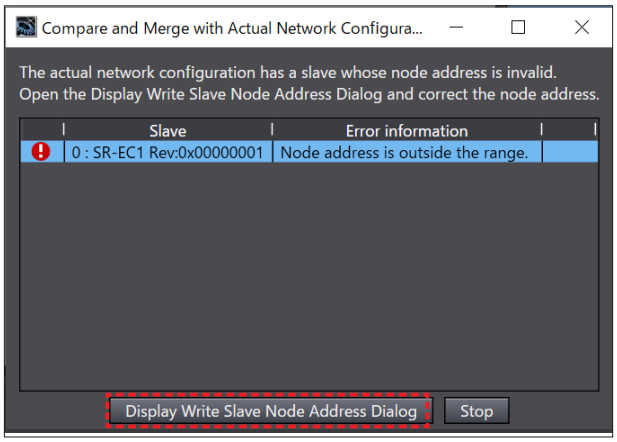
Click "Online" and right click the Master.



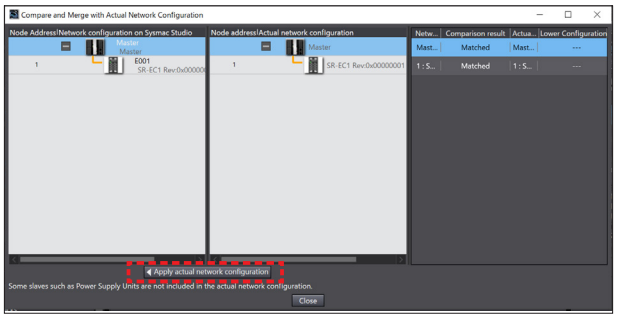
Select "Compare and Merge with Actual Network Configuration".



Click "Display Write Slave Node Address Dialog".



Click "Apply actual network configuration" and click "Apply".
Close this window after this operation.



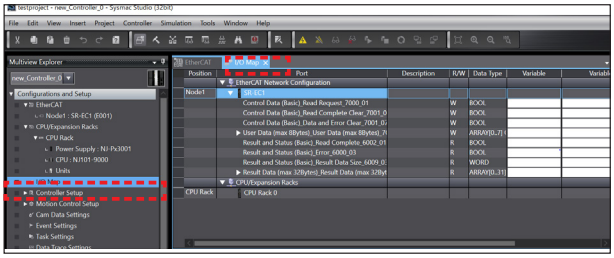
After completing the settings up to these points, return "Online" to "Offline".

Chapter 4 EXECUTING LADDER PROGRAMMES AND MONITORING READING DATA

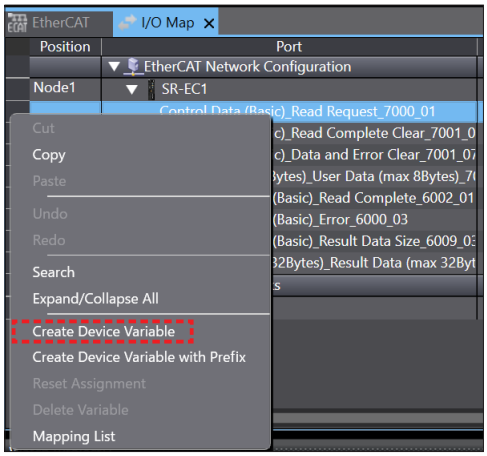
This chapter explains how to start SR reading through EtherCAT and how to check that the reading results are written to the data map.

1 Switch to online mode and start monitoring.

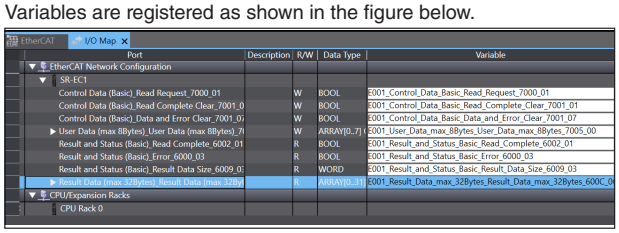
Double click "I/O Map" in "Multiview Explorer" and click "SR-EC1".



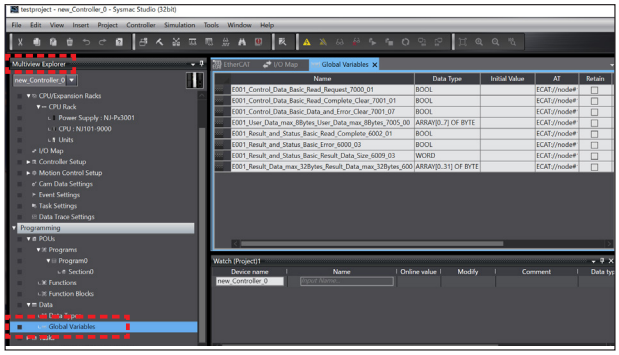
Right click on the variable in SR-EC1 and select "Create Device Variable".



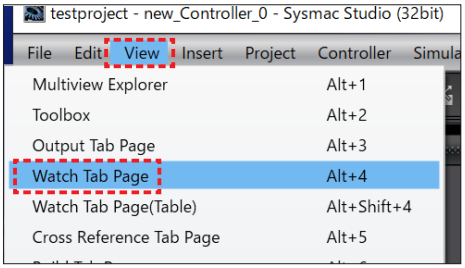
Variables are registered as shown in the figure below.



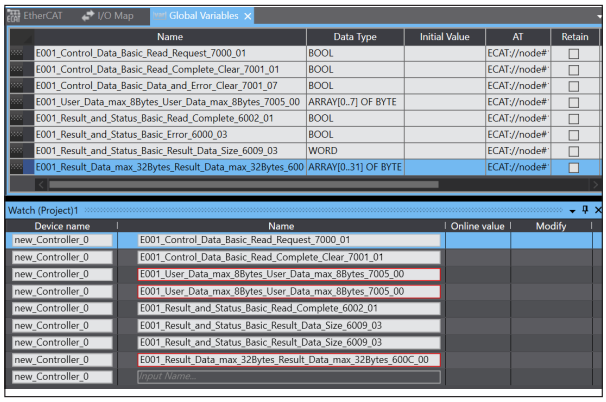
Double click "Global Variables" in "Multiview Explorer".



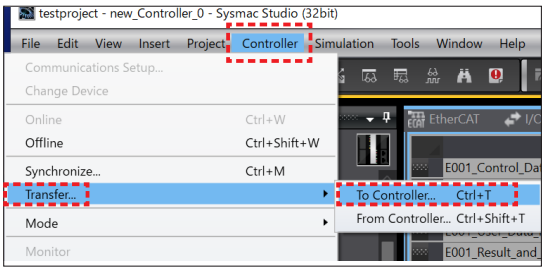
Click "View" → "Watch Tab Page".



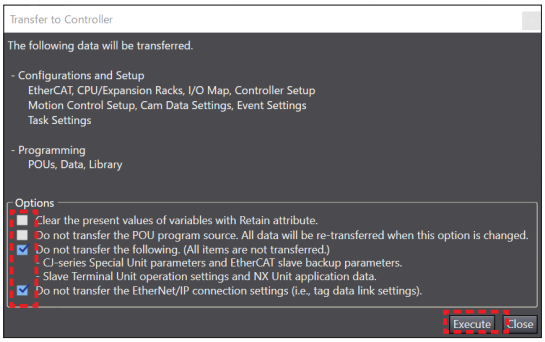
Drag and drop variables in Global Variables into the Watch Table.



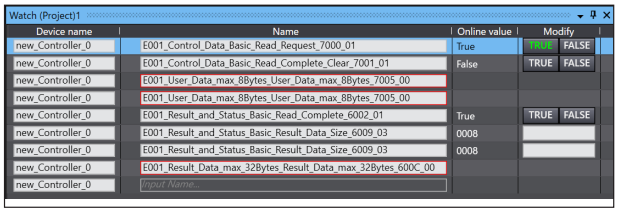
Click "Controller" → "Transfer" → "To Controller" .



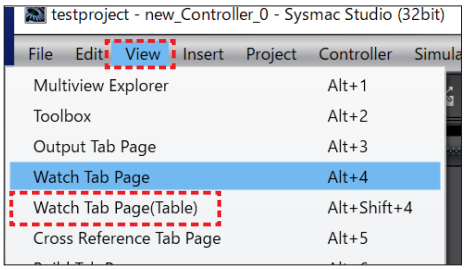
Check the boxes as shown below and click on "Execute".
Click "Operate" after this operation.



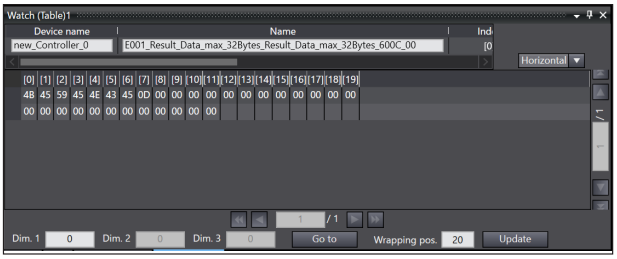
After returning to online again, click TRUE of Read Requset to start reading.



Click "View" → "Watch Tab Page(Table)".



You can check the read data is included when you entering the Result Data name in the Watch Table Name.



The read data is entered in hexadecimal, it must be converted to ASCII code.

4B 45 59 45 4E 43 45 0D

↓

KEYENCE[CR]

Sample code

KEYENCE

Specifications are subject to change without notice.

KEYENCE CORPORATION

1-3-14, Higashi-Nakajima, Higashi-Yodogawa-ku, Osaka, 533-8555, Japan PHONE: +81-6-6379-2211

www.keyence.com/glb

AUSTRIA
Phone: +43 (0)2236 378266 0

BELGIUM
Phone: +32 (0)15 281 222

BRAZIL
Phone: +55-11-3045-4011

CANADA
Phone: +1-905-366-7655

CHINA
Phone: +86-21-3357-1001

CZECH REPUBLIC
Phone: +420 220 184 700

FRANCE
Phone: +33 1 56 37 78 00

GERMANY
Phone: +49-6102-3689-0

HONG KONG
Phone: +852-3104-1010

HUNGARY
Phone: +36 1 802 7360

INDIA
Phone: +91-44-4963-0900

INDONESIA
Phone: +62-21-2966-0120

ITALY
Phone: +39-02-6688220

KOREA
Phone: +82-31-789-4300

MALAYSIA
Phone: +60-3-7883-2211

MEXICO
Phone: +52-55-8850-0100

NETHERLANDS
Phone: +31 (0)40 206 6100

PHILIPPINES
Phone: +63-(0)2-8981-5000

POLAND
Phone: +48 71 368 61 60

ROMANIA
Phone: +40 (0)269 232 808

SINGAPORE
Phone: +65-6392-1011

SLOVAKIA
Phone: +421 (0)2 5939 6461

SLOVENIA
Phone: +386 (0)1 4701 666

SWITZERLAND
Phone: +41 (0)43 455 77 30

TAIWAN
Phone: +886-2-2721-1080

THAILAND
Phone: +66-2-078-1090

UK & IRELAND
Phone: +44 (0)1908-696-900

USA
Phone: +1-201-930-0100

VIETNAM
Phone: +84-24-3772-5555

A4WW1-MAN-2062

Copyright (c) 2022 KEYENCE CORPORATION. All rights reserved. 193357GB 2122-1 D67GB Printed in Japan

D67GB-1